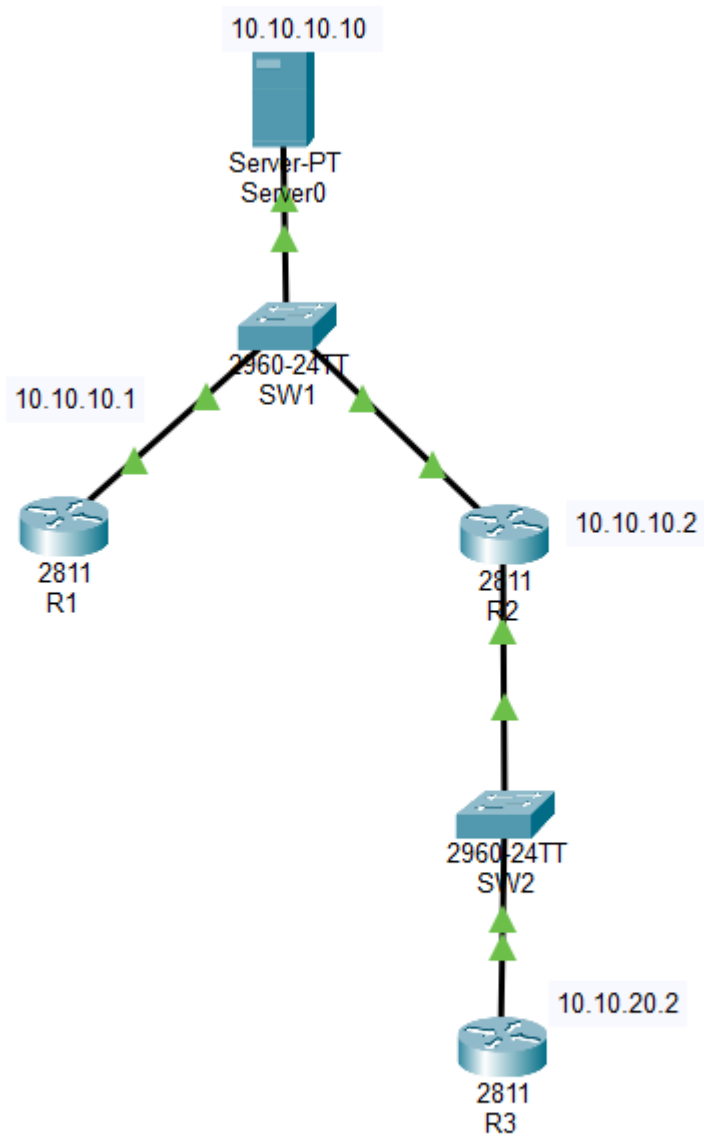




```
R2(config)#ip name-server 10.10.10.10
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R2#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R2#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R2#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R2#ping R1
Translating "R1"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

```

-----
R2#show arp
Protocol Address           Age (min)  Hardware Addr  Type   Interface
Internet 10.10.10.1             1    0090.2BD6.DD01  ARPA   FastEthernet0/0
Internet 10.10.10.2             -    0001.6429.4A01  ARPA   FastEthernet0/0
Internet 10.10.10.10           1    00D0.9761.4A29  ARPA   FastEthernet0/0
Internet 10.10.20.1            1    0060.2F93.2201  ARPA   FastEthernet0/1
Internet 10.10.20.2             -    0001.6429.4A02  ARPA   FastEthernet0/1
R2#clear arp-cache
R2#show arp
Protocol Address           Age (min)  Hardware Addr  Type   Interface
Internet 10.10.10.1             0    0090.2BD6.DD01  ARPA   FastEthernet0/0
Internet 10.10.10.2             -    0001.6429.4A01  ARPA   FastEthernet0/0
Internet 10.10.10.10           0    00D0.9761.4A29  ARPA   FastEthernet0/0
Internet 10.10.20.1            0    0060.2F93.2201  ARPA   FastEthernet0/1
Internet 10.10.20.2             -    0001.6429.4A02  ARPA   FastEthernet0/1
R2#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R2#show arp
Protocol Address           Age (min)  Hardware Addr  Type   Interface
Internet 10.10.10.1             4    0090.2BD6.DD01  ARPA   FastEthernet0/0
Internet 10.10.10.2             -    0001.6429.4A01  ARPA   FastEthernet0/0
Internet 10.10.10.10           4    00D0.9761.4A29  ARPA   FastEthernet0/0
Internet 10.10.20.1            4    0060.2F93.2201  ARPA   FastEthernet0/1
Internet 10.10.20.2             -    0001.6429.4A02  ARPA   FastEthernet0/1
R2#

```

An ARP entry maps an IP address to the corresponding MAC address of a device on the local network. A routing table tells the router where/how to forward an IP packet, but it does not provide the MAC address needed to create the local Ethernet frame

The routing table determines the next-hop IP address or outgoing interface, but Ethernet communication also requires a destination MAC address. ARP is therefore used to discover the MAC address associated with the next-hop IP address so that the router can create and send the Ethernet frame.